Azure Enterprise Infrastructure

Capstone Project — Deployment Screenshots

Step 1 — Virtual Network & Subnets

10.0.0.0/16 Delete address space

10.0.0.0 /16

10.0.0.0 - 10.0.255.255 65,536 addresses

Subnets	IP address range	Size	NAT gateway
WebSubnet	10.0.1.0 - 10.0.1.255	/24 (256 addresses)	-
AppSubnet	10.0.2.0 - 10.0.2.255	/24 (256 addresses)	-
DbSubnet	10.0.3.0 - 10.0.3.255	/24 (256 addresses)	-

Home

vnet-capstone-1776582075267 | Overview

Deployment

Search

Overview

Inputs

Outputs

Template

Delete Cancel Redeploy Download Refresh

Your deployment is complete

Deployment name : vnet-capstone-1776582075267

Subscription : PAYG-Sandboxes

Resource group : rg_sb_eastus_335199_1_177658108963

Start time : 4/19/2026, 11:01:26 AM

Correlation ID : 7d89039d-bcda-4e80-be95-7b3b8d001f9a

Deployment details

Next steps

Go to resource

Step 2 — Network Security Group (nsg-web)

Home › CreateNetworkSecurityGroupBladeV2-20260419110226 | Overview › nsg-web

nsg-web | Subnets

Network security group

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer

+ Associate

Search subnets

Name	Address range	Virtual network
WebSubnet	10.0.1.0/24	vnet-capstone

Give feedback

Home

CreateNetworkSecurityGroupBladeV2-20260419110226 | Overview

Deployment

Search

- Overview
- Inputs
- Outputs
- Template

Delete | Cancel | Redeploy | Download | Refresh

Your deployment is complete

Deployment name : CreateNetworkSecurityGroupBladeV2-20260419110226
Subscription : PAYG-Sandboxes
Resource group : rg_sb_eastus_335199_1_177658108963
Start time : 4/19/2026, 11:06:34 AM
Correlation ID : 088dc24a-dbd1-47be-8f98-9f765a1c6219

Deployment details

Next steps

Go to resource

Step 3 — NSG Inbound Security Rules

Home › CreateNetworkSecurityGroupBladeV2-20260419110226 | Overview › nsg-web

nsg-web | Inbound security rules

Network security group

Search

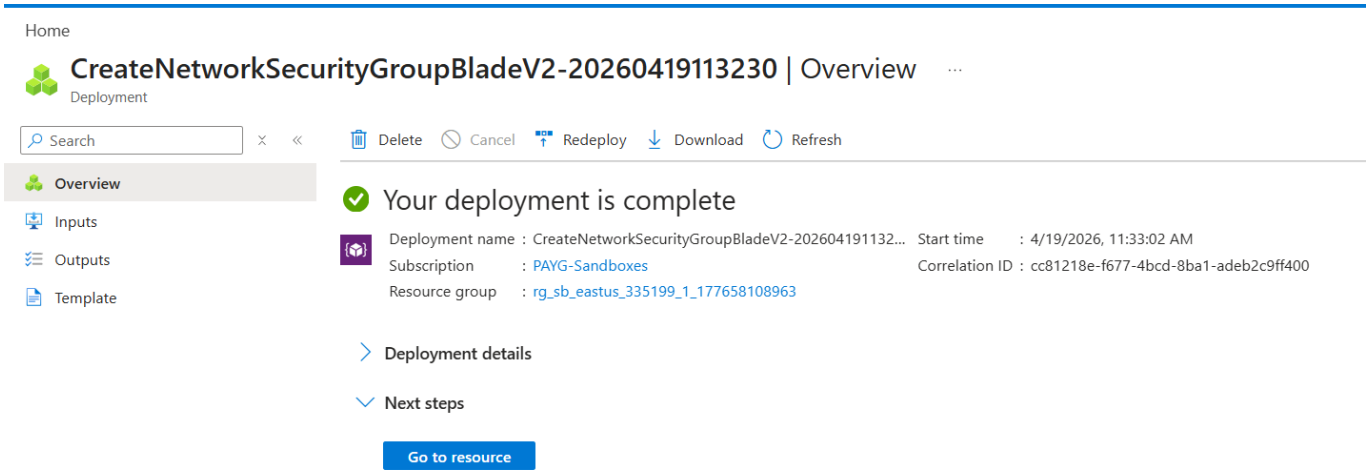
- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
  - Inbound security rules
  - Outbound security rules
  - Network interfaces
  - Subnets
  - Properties
  - Locks

+ Add | Hide default rules | Refresh | Delete | Give feedback

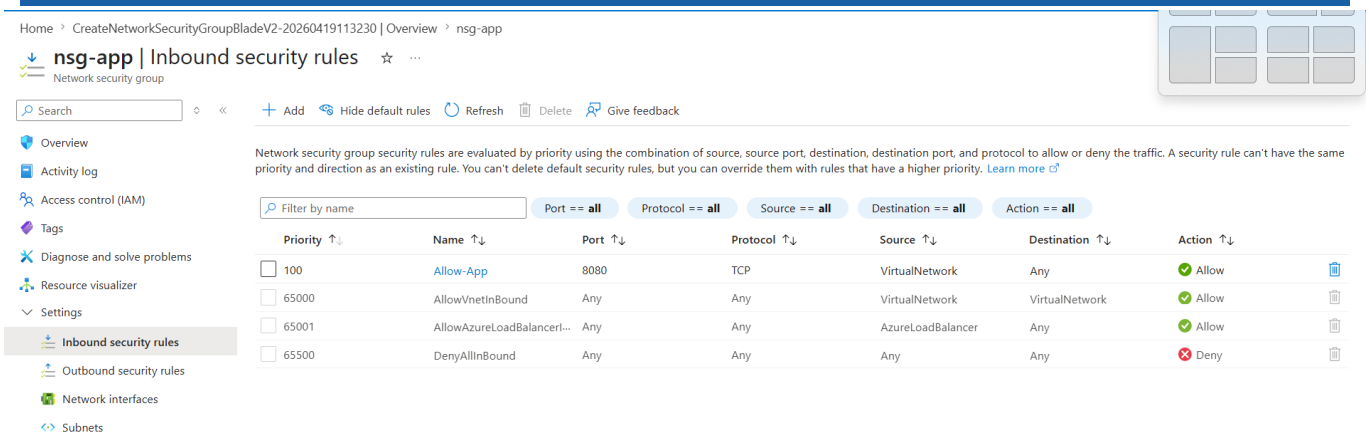
Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. Learn more

Filter by name | Port == all | Protocol == all | Source == all | Destination == all | Action == all

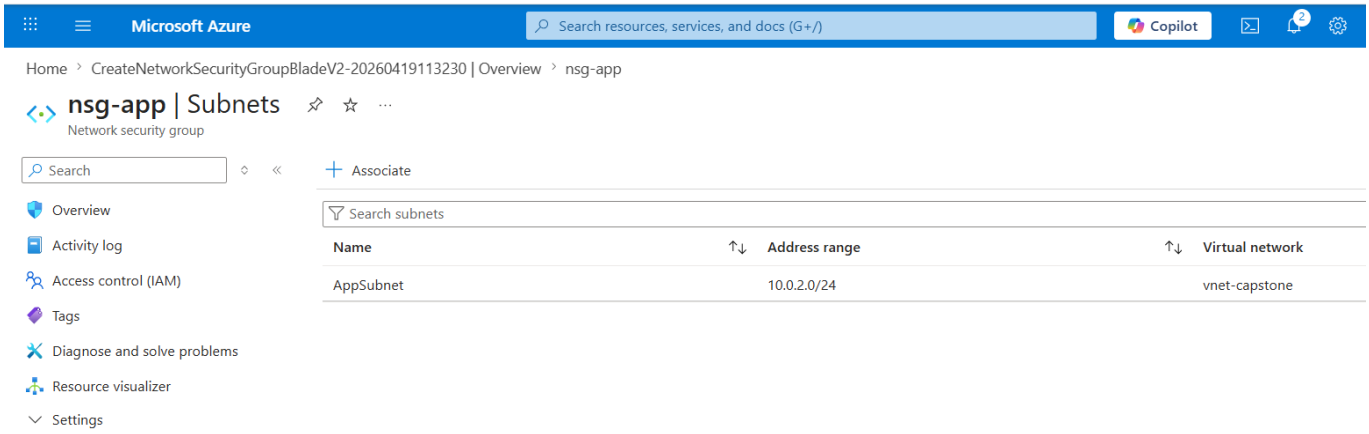
Priority	Name	Port	Protocol	Source	Destination	Action
100	Allow-LB-Probe	80	TCP	AzureLoadBalancer	Any	Allow
200	Allow-HTTP	80	TCP	AzureLoadBalancer	Any	Allow
300	Allow-HTTPS	443	TCP	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerI···	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny



Step 4 — NSG Subnet Association



Step 5 — Virtual Machine 1 (vm-web-01)



Home › CreateNetworkSecurityGroupBladeV2-20260419113936 | Overview › nsg-db

nsg-db | Inbound security rules

Network security group

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
  - Inbound security rules
  - Outbound security rules
  - Network interfaces
  - Subnets
  - Properties

Add | Hide default rules | Refresh | Delete | Give feedback

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority.
Learn more

Filter by name

Port == **all** | Protocol == **all** | Source == **all** | Destination == **all** | Action == **all**

Priority	Name	Port	Protocol	Source
100	Allow-SQL	1433	TCP	VirtualNetwork
65000	AllowVnetInBound	Any	Any	VirtualNetwork
65001	AllowAzureLoadBalan...	Any	Any	AzureLoadBalance
65500	DenyAllInBound	Any	Any	Any

Step 6 — Load Balancer Configuration

Microsoft Azure | Search resources, services, and docs (G+/) | Copilot

Home

nat-capstone-20260419115358-90 | Overview

Deployment

Search

- Overview
- Inputs
- Outputs
- Template

Delete | Cancel | Redeploy | Download | Refresh

Your deployment is complete

Deployment name : nat-capstone-20260419115358-90
Subscription : PAYG-Sandboxes
Resource group : rg_sb_eastus_335199_1_177658108963

Start time : 4/19/2026, 11:55:24 AM
Correlation ID : 2e5bb024-51bc-42fc-816a-f64491ef34ea

Deployment details

Next steps

Go to resource

Home

CreateVm-canonical.ubuntu-24_04-lts-server-20260419115718 | Overview

Deployment

Search

Overview
Inputs
Outputs
Template

Delete Cancel Redeploy Download Refresh

Your deployment is complete

Deployment name : CreateVm-canonical.ubuntu-24_04-lts-server-20260419115718
Subscription : PAYG-Sandboxes
Resource group : rg_sb_eastus_335199_1_177658108963
Start time : 4/19/2026, 12:15:07 PM
Correlation ID : 8a73c944-b37c-4fb5-ab7a-f971edff5eb1

Deployment details

Next steps

Set up auto-shutdown Recommended
Monitor VM health, performance, and network dependencies Recommended
Run a script inside the virtual machine Recommended

Go to resource | Create another VM | Scale out your VM

Step 7 — Virtual Machine 2 (vm-web-02)

Home

CreateVm-canonical.ubuntu-24_04-lts-server-20260419121629 | Overview

Deployment

Search

Overview
Inputs
Outputs
Template

Delete Cancel Redeploy Download Refresh

Deployment is in progress

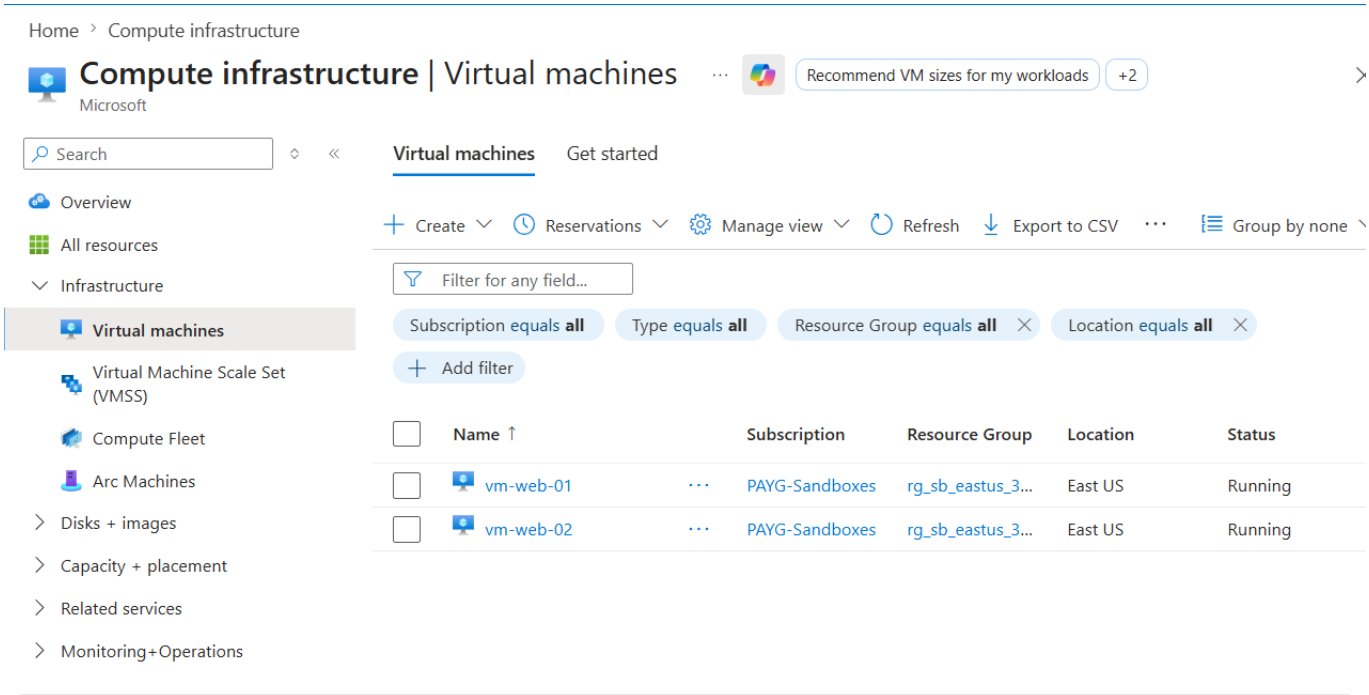
Deployment name : CreateVm-canonical.ubuntu-24_04-lts-server-202604...
Subscription : PAYG-Sandboxes
Resource group : rg_sb_eastus_335199_1_177658108963
Start time : 4/19/2026, 12:23:38 PM
Correlation ID : eb9cb367-bbc4-4522-8b32-0aae2838edfd

Deployment details

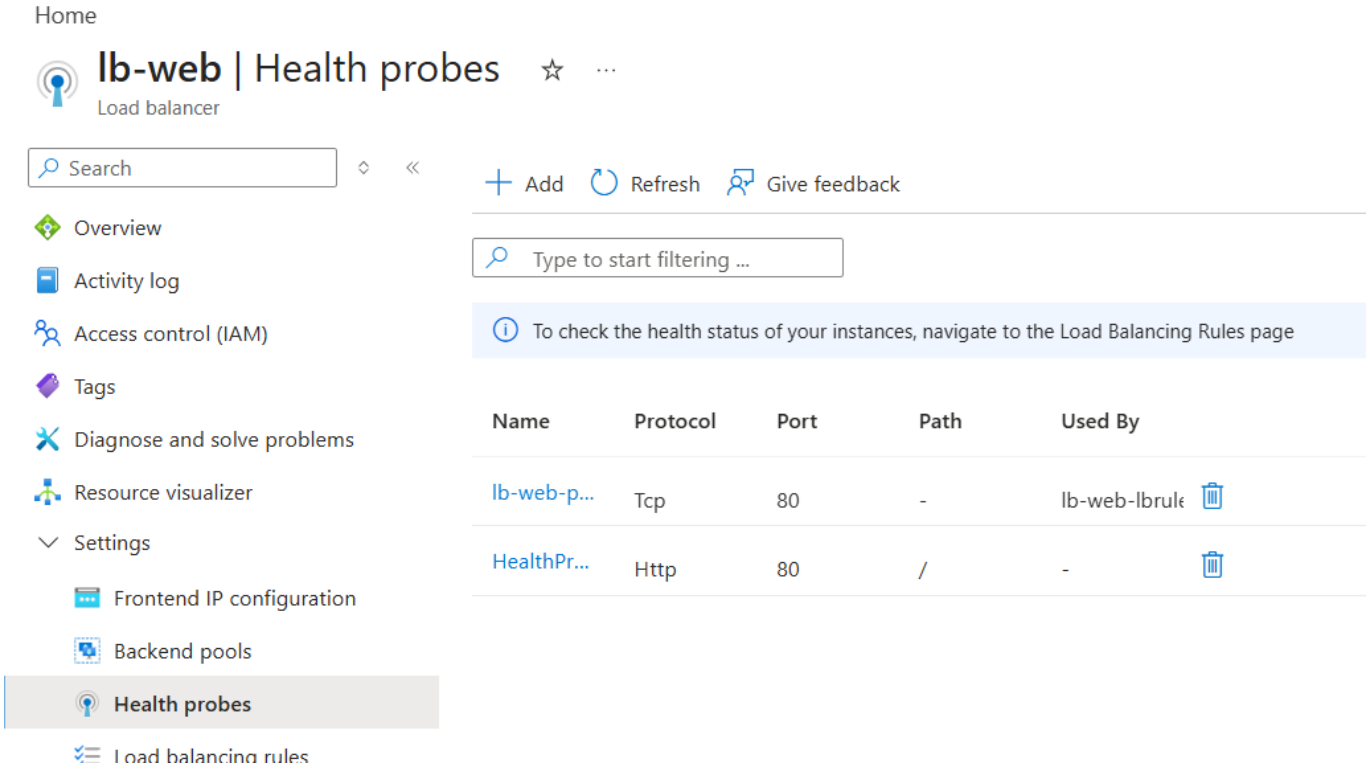
Resource	Type	Status	Operation details
vm-web-02182_z1	Microsoft.Network/networkInter	Created	Operation details
vm-web-02-ip	Public IP address	OK	Operation details
vm-web-02-nsg	Network security group	OK	Operation details

Next steps

Step 8 — Nginx Working (Load Balancer Test)



Step 9 — Azure SQL Database



Microsoft Azure | Search resources, services, and docs (G+/) | Copilot

Home > Azure SQL | SQL databases

Create SQL Database

Microsoft

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. Learn more

Connectivity method *

- ○ No access
- ○ Public endpoint
- ◉ Private endpoint

Private endpoints

Private endpoint connections are associated with a private IP address within a Virtual Network. The list below shows all the private endpoint connections for this server. Note that private endpoint connections are defined at the server level and they provide access to all databases in the server. Learn more

+ Add private endpoint

Name	Subscription	Resource group	Region	Subnet
pe-sql	PAYG-Sandboxes	rg_sb_eastus_335199_1...	eastus	vnet-capstone / DbSubnet

Connection policy

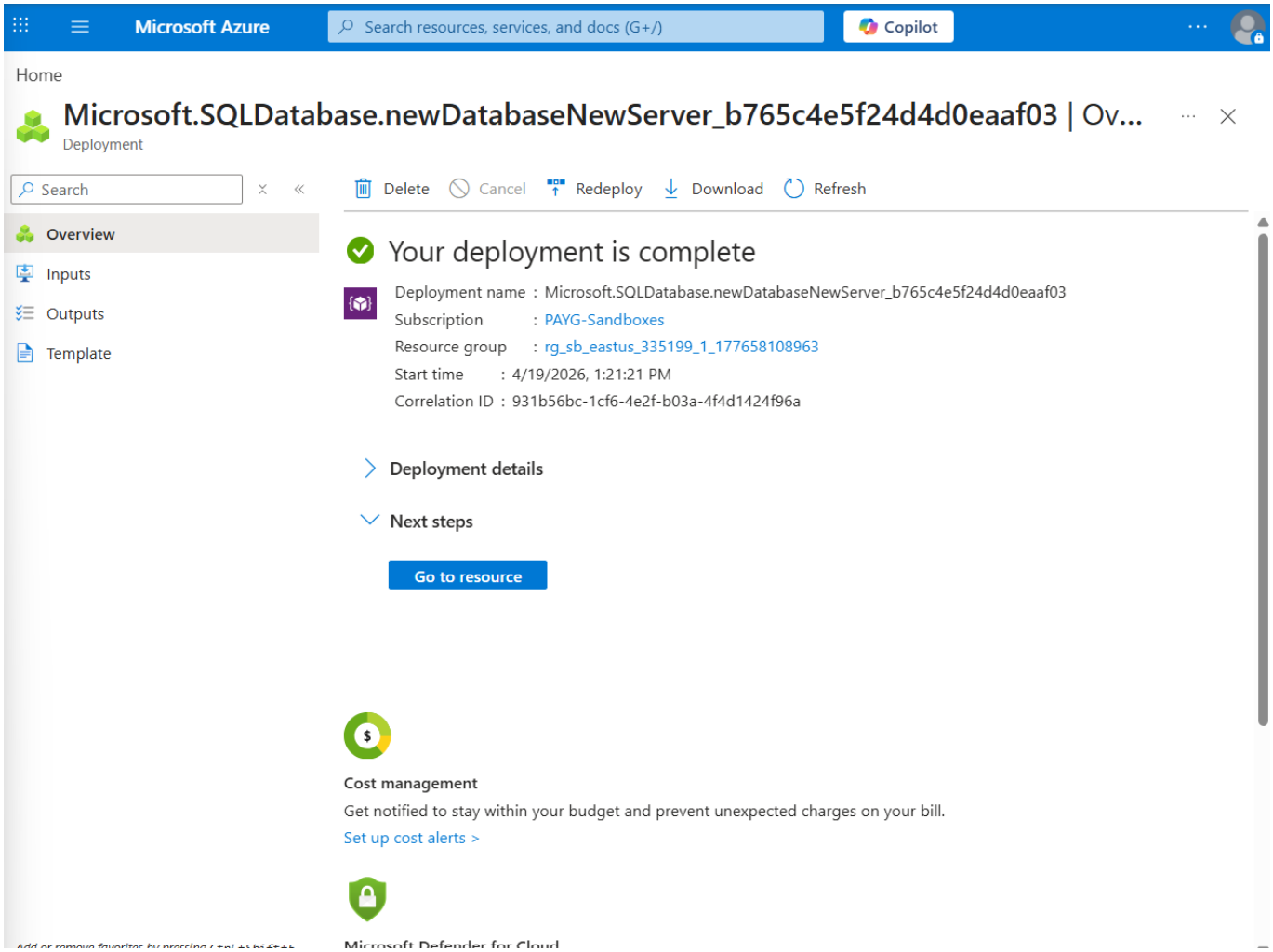
Configure how clients communicate with your SQL database server. Learn more

Connection policy

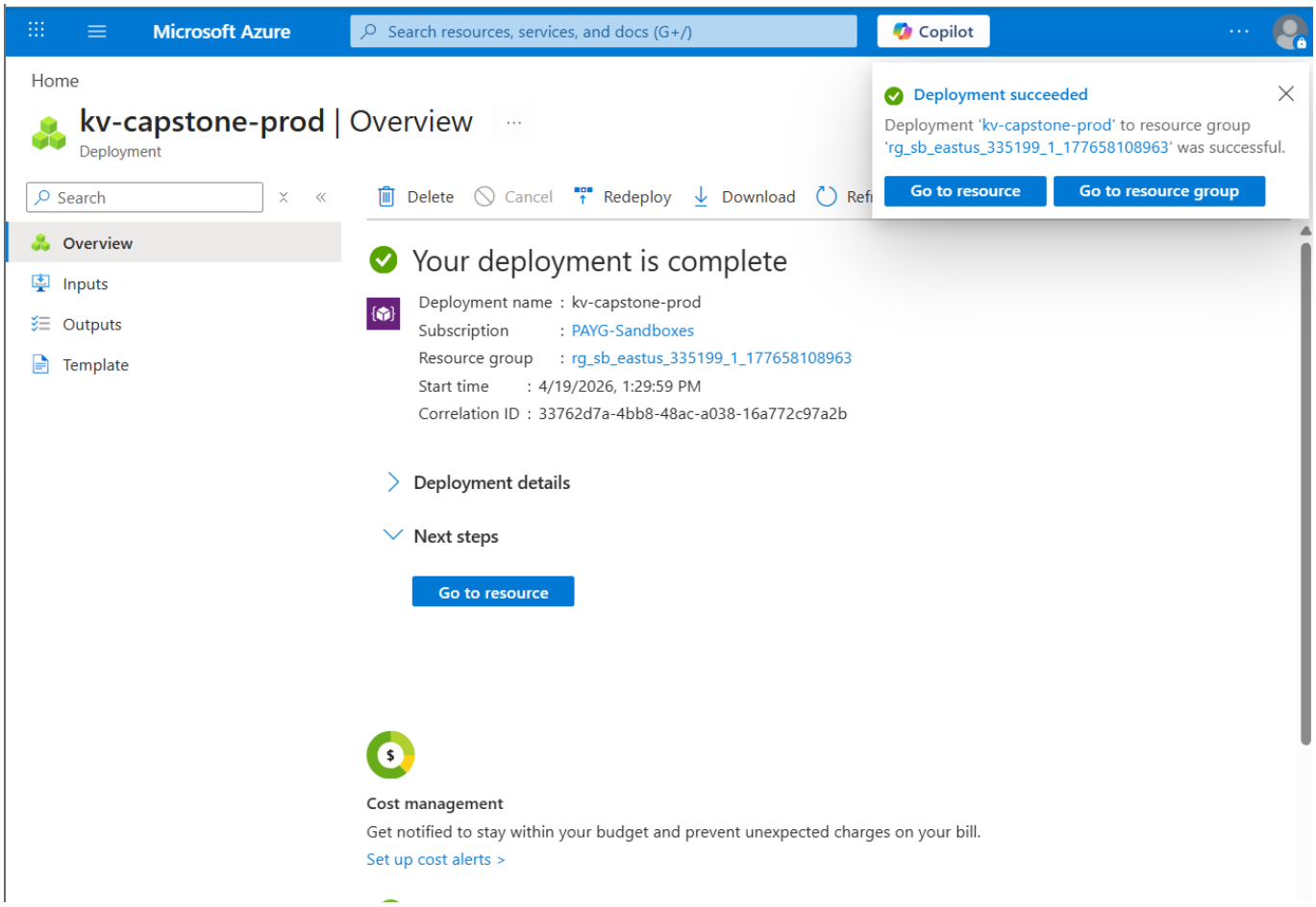
- ◉ Default - Uses Redirect policy for all client connections originating inside of Azure (except Private Endpoint connections) and Proxy for all client connections originating outside Azure
- ○ Proxy - All connections are proxied via the Azure SQL Database gateways
- ○ Redirect - Clients establish connections directly to the node hosting the database

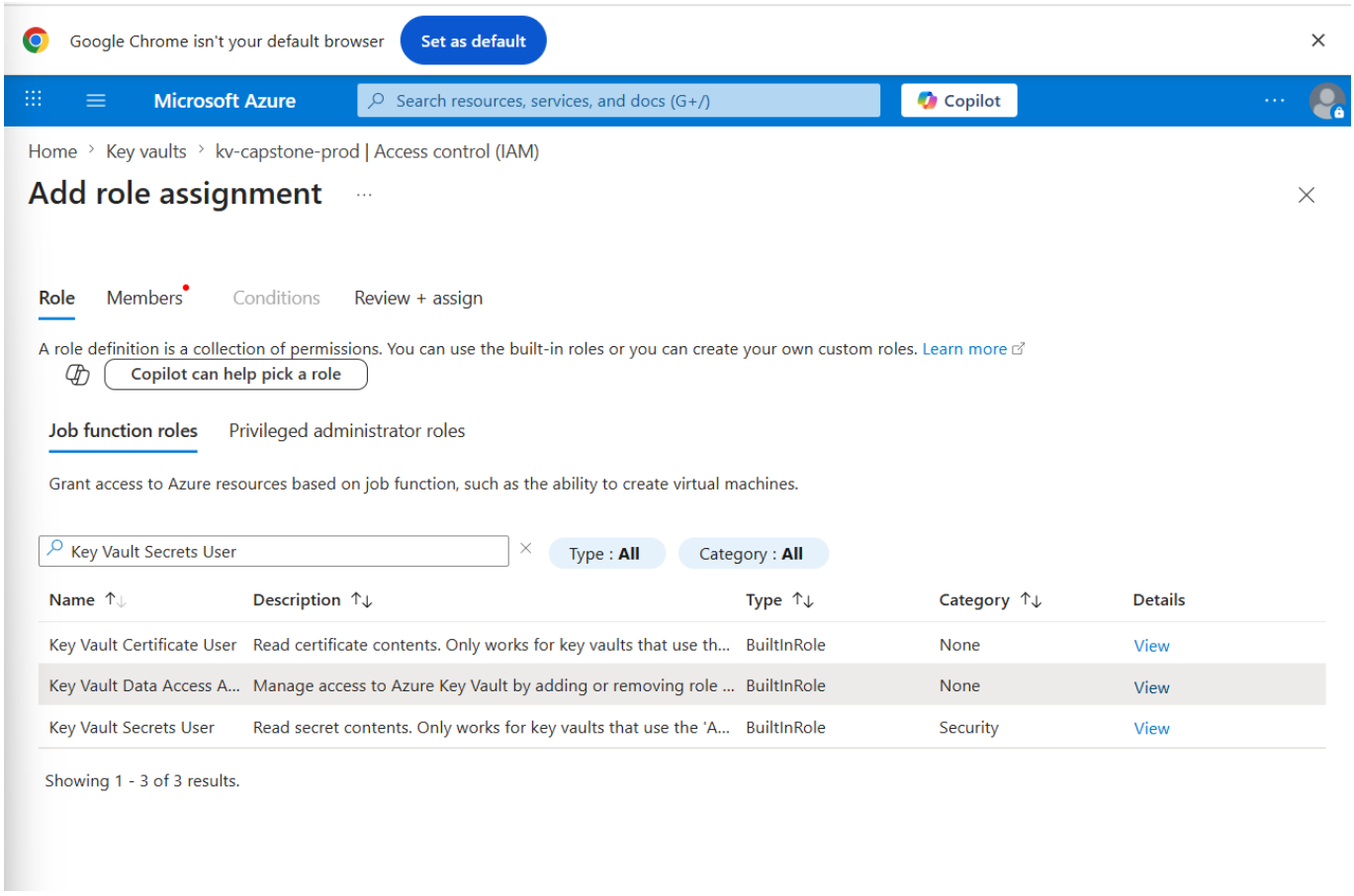
Review + create | < Previous | Next : Security >

Step 10 — SQL Networking & Private Endpoint

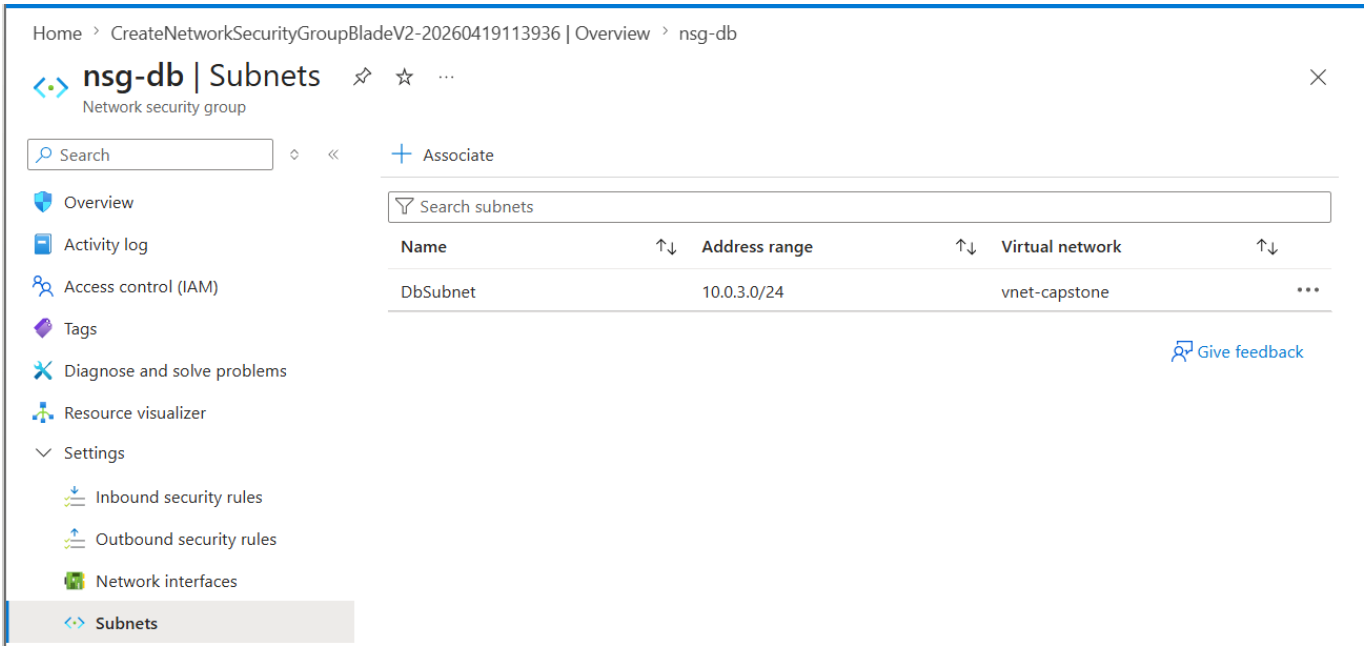


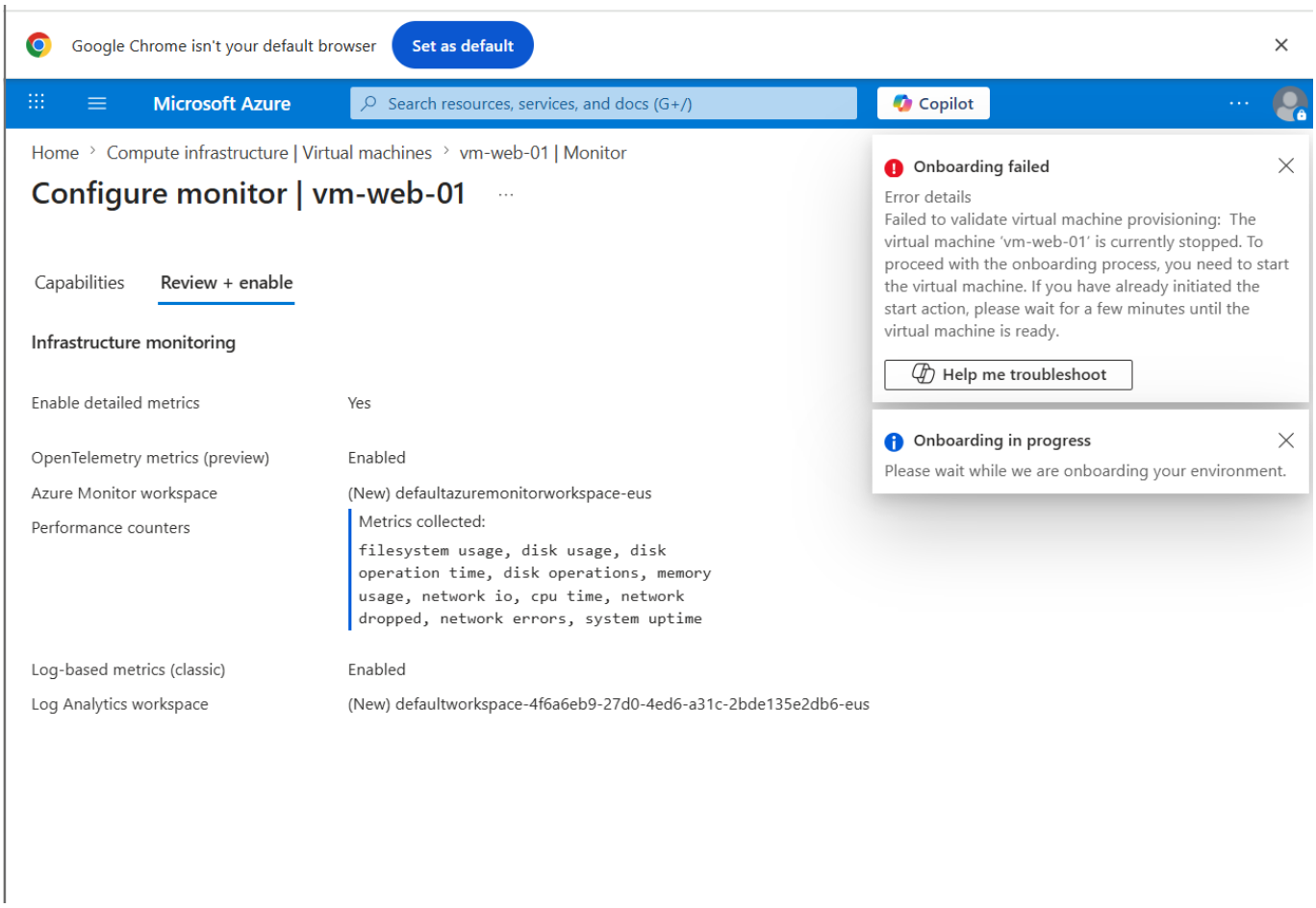
Step 11 — Monitoring (Azure Monitor)





Step 12 — CPU Alert Rule





Step 13 — Key Vault & Secrets

Microsoft Azure | Search resources, services, and docs (G+/) | Copilot

Home > Monitor | Alerts

Create an alert rule

Scope | Condition | Actions | Details | Tags | Review + create

Create an alert rule to identify and address issues when important conditions are found in your monitoring data. Learn more

Scope level: Subscription

+ Select scope

Resource	Hierarchy
rg_sb_eastus_335199_1_177659455850	PAYG-Sandboxes